All-Corner Post-Route STA Analysis

Static timing analysis across 16 PVT corners using routed design data, timing constraints, extracted parasitics, and corner-specific Liberty libraries.

Executive Summary

Corners analyzed: 16
Corners with negative setup WNS: 6
Corners with negative hold WNS: 0
Worst setup WNS: -30.7252 ns at ss_n40C_1v28.

Timing Metrics by Corner

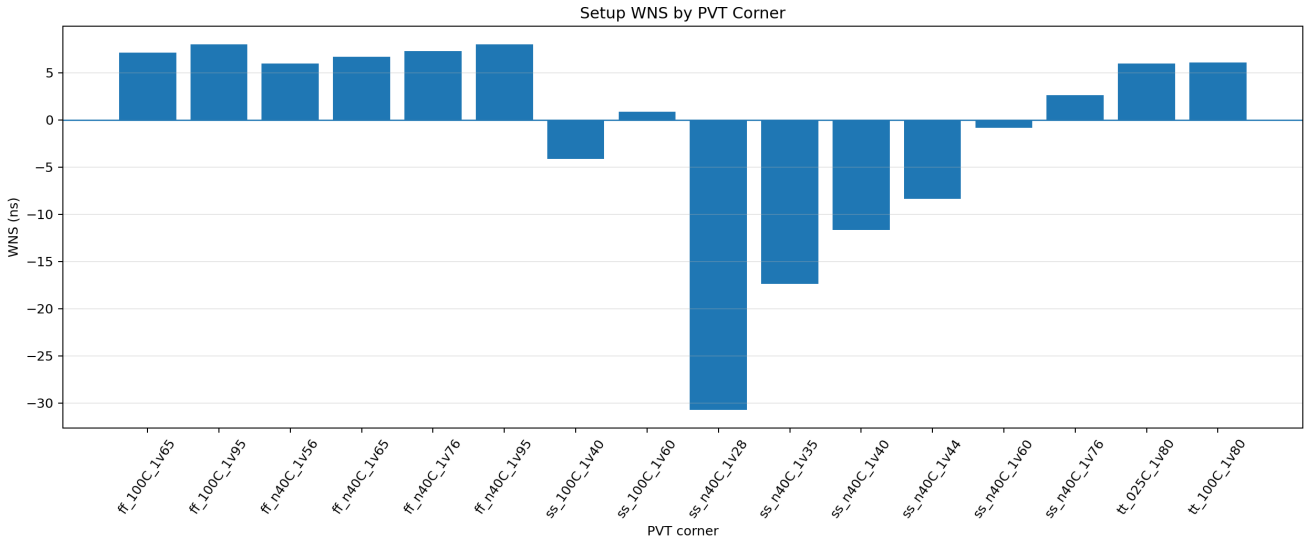
Corner	Setup WNS (ns)	Setup TNS (ns)	Hold WNS (ns)	Hold TNS (ns)
ff_100C_1v65	7.1134	0.0000	0.2508	0.0000
ff_100C_1v95	8.0138	0.0000	0.1985	0.0000
ff_n40C_1v56	5.9507	0.0000	0.2916	0.0000
ff_n40C_1v65	6.6615	0.0000	0.2570	0.0000
ff_n40C_1v76	7.2816	0.0000	0.2274	0.0000
ff_n40C_1v95	7.9913	0.0000	0.1916	0.0000
ss_100C_1v40	-4.1254	-398.1285	0.9008	0.0000
ss_100C_1v60	0.8405	0.0000	0.6404	0.0000
ss_n40C_1v28	-30.7252	-13550.8096	1.7552	0.0000
ss_n40C_1v35	-17.3561	-6161.1196	1.3135	0.0000
ss_n40C_1v40	-11.6665	-3219.5767	1.0999	0.0000
ss_n40C_1v44	-8.3726	-1788.6748	0.9676	0.0000
ss_n40C_1v60	-0.8504	-17.4777	0.6539	0.0000
ss_n40C_1v76	2.6041	0.0000	0.4871	0.0000
tt_025C_1v80	5.9879	0.0000	0.3111	0.0000
tt_100C_1v80	6.0693	0.0000	0.3156	0.0000

Interpretation and Limitations

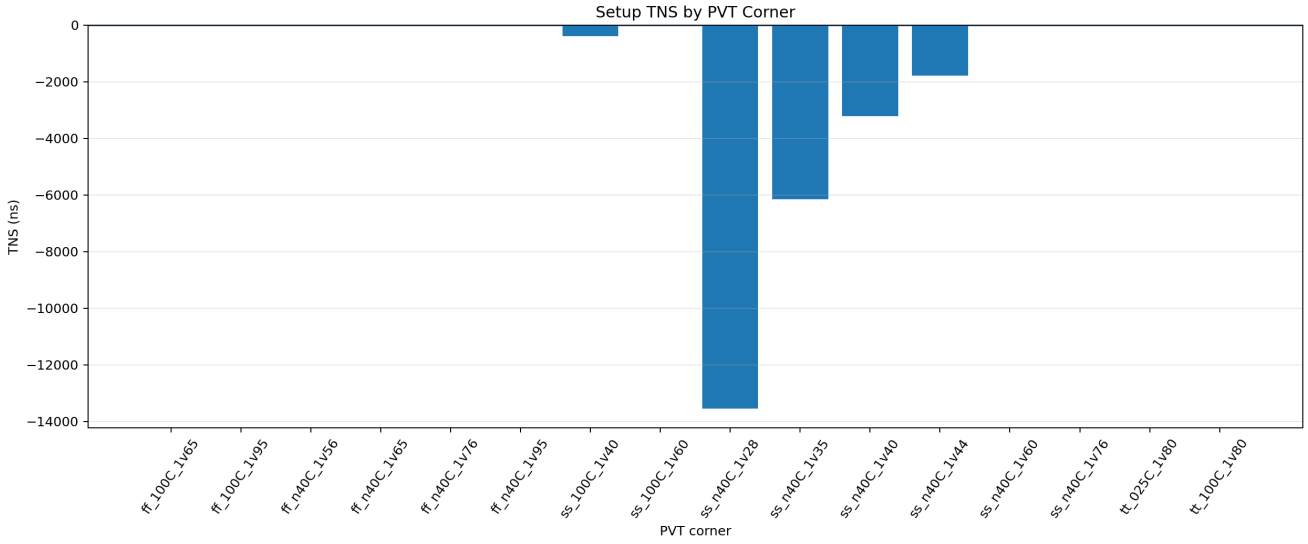
Negative setup WNS indicates setup timing violations under the analyzed constraints. Negative hold WNS indicates hold timing violations. The detailed corner reports should be reviewed to investigate failing paths. Zero TNS alone does not prove timing closure if WNS is negative. Results are specific to the loaded design, constraints, Liberty libraries, and parasitics; they are not a substitute for a complete sign-off review.

Timing Graphs

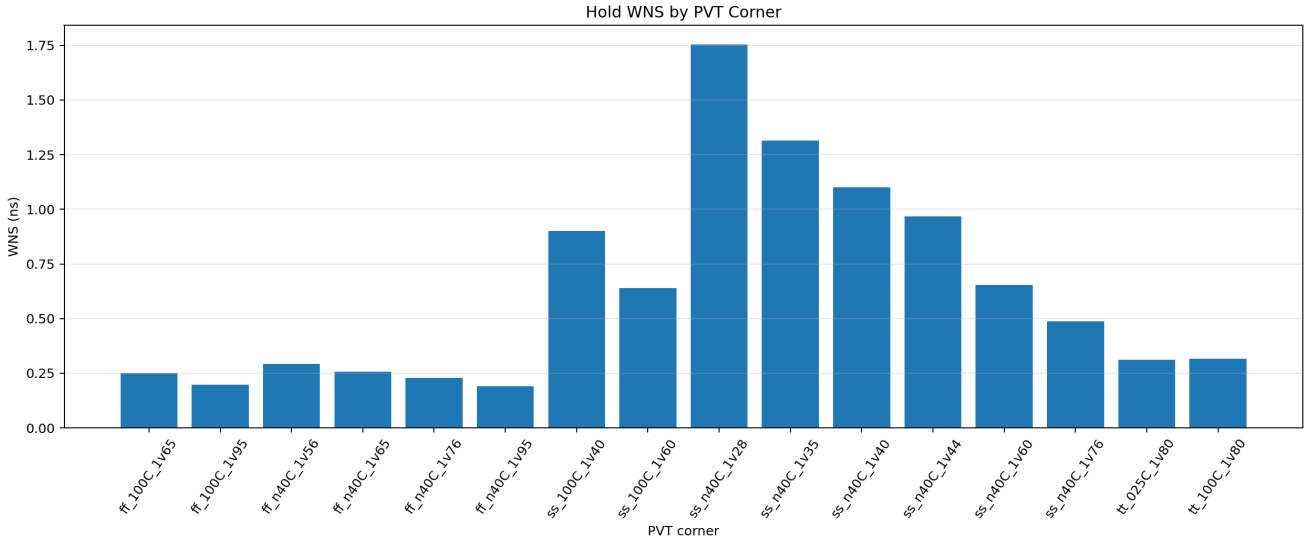
Setup WNS by PVT Corner



Setup TNS by PVT Corner



Hold WNS by PVT Corner



Hold TNS by PVT Corner

